

2. Pandas Data Structures

In Pandas, there are 2 main data structures:

1. Series → Single Column
2. DataFrame → Table (Rows + Columns)

1. Series in Pandas

A Series is:

- One-dimensional data
- Like a single column in Excel

Example:

Index	Marks
0	85
1	90
2	88

Creating Series

First import Pandas:

```
import pandas as pd
```

Example 1: Simple Series

```
marks = pd.Series([85, 90, 88, 75])  
  
print(marks)
```

Output

```
0  85  
1  90  
2  88
```

3 75
dtype: int64

Understanding the Output

Part	Meaning
0,1,2,3	Index
85,90,88	Values
dtype	Data type

Example 2: Series with Custom Index

```
students = pd.Series(  
    [85, 90, 88],  
    index=["Rahul", "Aman", "Priya"]  
)  
  
print(students)
```

Output

```
Rahul 85  
Aman 90  
Priya 88  
dtype: int64
```

Indexing in Series

Indexing means accessing data.

Example 1: Access by Position

```
marks = pd.Series([85, 90, 88])  
  
print(marks[0])
```

Output:

85

Example 2: Access by Custom Index

```
students = pd.Series(  
    [85, 90, 88],  
    index=["Rahul", "Aman", "Priya"]  
)  
  
print(students["Aman"])
```

Output:

90

Multiple Values Access

```
print(students[["Rahul", "Priya"]])
```

Output:

Rahul 85
Priya 88

Operations on Series

We can perform:

- Addition
- Multiplication
- Mean
- Maximum
- Minimum

Example 1: Addition

```
marks = pd.Series([85, 90, 88])  
  
print(marks + 5)
```

Output:

```
0    90  
1    95  
2    93
```

Example 2: Multiplication

```
print(marks * 2)
```

Output:

```
0    170  
1    180  
2    176
```

Example 3: Mean (Average)

```
print(marks.mean())
```

Output:

```
87.66
```

Example 4: Maximum Value

```
print(marks.max())
```

Output:

90

Example 5: Minimum Value

```
print(marks.min())
```

Output:

85

2. DataFrame in Pandas

A DataFrame is:

- Two-dimensional data
 - Rows + Columns
 - Similar to Excel table
-

Example of DataFrame

Name	Age	Marks
Rahul	20	85
Aman	21	90
Priya	19	88

Creating DataFrame

Example 1: DataFrame from Dictionary

```
import pandas as pd
```

```
data = {
```

```
"Name": ["Rahul", "Aman", "Priya"],  
"Age": [20, 21, 19],  
"Marks": [85, 90, 88]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

Output

	Name	Age	Marks
0	Rahul	20	85
1	Aman	21	90
2	Priya	19	88

Understanding DataFrame Parts

Part	Meaning
Rows	Horizontal data
Columns	Vertical data
Index	Row numbers

Rows and Columns

Get Column Names

```
print(df.columns)
```

Output:

```
Index(['Name', 'Age', 'Marks'], dtype='object')
```

Get Rows

```
print(df.index)
```

Output:

```
RangeIndex(start=0, stop=3, step=1)
```

Select Single Column

```
print(df["Name"])
```

Output:

```
0  Rahul
1  Aman
2  Priya
```

Select Multiple Columns

```
print(df[["Name", "Marks"]])
```

Select Row using loc[]

```
print(df.loc[1])
```

Output:

```
Name  Aman
Age    21
Marks  90
```

Select Row using iloc[]

```
print(df.iloc[0])
```

Output:

```
Name  Rahul
Age    20
Marks  85
```

DataFrame from List

Example

```
data = [  
    ["Rahul", 20, 85],  
    ["Aman", 21, 90],  
    ["Priya", 19, 88]  
]  
  
df = pd.DataFrame(  
    data,  
    columns=["Name", "Age", "Marks"]  
)  
  
print(df)
```

DataFrame from Dictionary

Example

```
data = {  
    "Name": ["Rahul", "Aman"],  
    "Marks": [85, 90]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

DataFrame from CSV File

CSV = Comma Separated Values

Example CSV file:

students.csv

Name	Age	Marks
Rahul	20	85
Aman	21	90

Read CSV File

```
df = pd.read_csv("students.csv")  
  
print(df)
```

DataFrame from Excel File

Install Excel support:

```
pip install openpyxl
```

Read Excel File

```
df = pd.read_excel("students.xlsx")  
  
print(df)
```

DataFrame from SQL Database

Suppose data is stored in MySQL.

Install SQLAlchemy

```
pip install sqlalchemy
```

Example: Read SQL Table

```
from sqlalchemy import create_engine  
import pandas as pd
```

```
engine = create_engine(
    "mysql+pymysql://root:password@localhost/school"
)

df = pd.read_sql(
    "SELECT * FROM students",
    engine
)

print(df)
```

Real-Life Example

Suppose you have:

- Student records
- Sales data
- Employee data
- Shop transactions

Pandas DataFrame helps:

- Analyze reports
- Filter records
- Generate summaries
- Export Excel reports

Difference Between Series and DataFrame

Feature	Series	DataFrame
Dimension	1D	2D
Structure	Single column	Multiple columns
Example	Marks	Student table

Summary

Topic	Explanation
Series	Single column
DataFrame	Table data
Indexing	Access data
loc[]	Access row by label
iloc[]	Access row by position
read_csv()	Read CSV
read_excel()	Read Excel
read_sql()	Read SQL table